

Satyam Awasthi

Researcher & Software Engineer

✉ satyam.aw@yahoo.com — [satyam-aw.github.io](https://github.com/satyam-aw) — [satyam-aw](https://www.linkedin.com/in/satyam-aw) — [0000-0002-6095-4303](https://orcid.org/0000-0002-6095-4303) — [in](https://www.linkedin.com/in/satyam-aw) LinkedIn

Professional Summary

Computer Science researcher focused on **Autonomous Systems**, **Spatial Perception**, and **Model-Based and Learning-Based Control**. My research explores how perception, predictive modeling, and planning can be integrated with control frameworks to enable robust long-horizon autonomy in robotic systems operating under uncertainty.

Research Interests

Autonomous Systems, Spatial Perception, Model-Based and Learning-Based Control, and Robot Planning, with emphasis on long-horizon decision-making under uncertainty in real-world robotic environments.

Education

University of California, Santa Barbara (UCSB)
M.S. in Computer Science (GPA: 3.91/4.00)

Santa Barbara, CA
Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur)
B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0)

Kharagpur, India
Jul 2016 – Jul 2020

Research Experience

Four Eyes Laboratory, University of California, Santa Barbara
Graduate Researcher

Sep 2021 – Jun 2023
Advisor: [Prof. Tobias Höllerer](#), Co-Advisor: [Prof. Michael Beyeler](#)

- Developed **EyeTTS**, an open-source framework for evaluating and calibrating eye tracking during mixed-reality locomotion, resulting in first-author publications at IEEE ISMAR Adjunct and IEEE VR Workshops.
- Designed experimental protocols and software pipelines to quantify gaze-tracking drift during active user motion.
- Curated a multi-device eye-tracking dataset and conducted user studies across diverse locomotion layouts.
- Published first-author poster papers at IEEE VR and IEEE ISMAR Adjunct venues.

Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur
Undergraduate Researcher

Jan 2016 – Dec 2018

- Contributed to the development of **Eklavya 6.0**, the IIT Kharagpur autonomous ground vehicle that finished **Runner-Up** at the **Intelligent Ground Vehicle Competition (IGVC) 2018**. [Competition Video](#) · [Team AGV](#)
- Developed a hardware-agnostic navigation stack integrating EKF-based localization from LiDAR, IMU, and GPS streams.
- Implemented visual obstacle segmentation using a custom U-Net architecture and hybrid motion planning using A*/Dijkstra and Dynamic Window Approach.
- Designed decentralized multi-agent coordination using relative-pose estimation over an ad-hoc mesh network.

Publications

Peer-Reviewed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing
Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.
COMSNETS, 2022. DOI: [10.1109/COMSNETS53615.2022.9668531](https://doi.org/10.1109/COMSNETS53615.2022.9668531)

Workshop & Poster Publications

Eye Tracking Performance in Mobile Mixed Reality
Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias Höllerer.
IEEE VRW, 2024. DOI: [10.1109/VRW62533.2024.00321](https://doi.org/10.1109/VRW62533.2024.00321)

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion

Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias Höllerer.

ISMAR Adjunct, 2023. DOI: [10.1109/ISMAR-Adjunct60411.2023.00104](https://doi.org/10.1109/ISMAR-Adjunct60411.2023.00104)

Research Artifacts

EyeTTS Calibration Framework

2022–2024

[Calibration Framework](#) · [User Study Framework](#)

- Developed an extensible framework for evaluating and calibrating eye-tracking systems during mixed-reality locomotion.
- Implemented device-agnostic benchmarking pipelines and automated drift-correction procedures.
- Released reproducible software infrastructure supporting eye-tracking research.

EyeTTS Mixed-Reality Eye Tracking Dataset

2022–2024

[Dataset](#) · [User Study Videos](#)

- Curated a multi-device mixed-reality locomotion dataset for calibration and performance evaluation.
- Collected gaze trajectories, calibration measurements, and locomotion metadata across diverse MR layouts.
- Supported experimental validation reported in ISMAR and VRW publications.

Selected Projects

JitterNot

Jan 2022 – Mar 2022

Learning-Based Model Predictive Control for Adaptive Video Streaming (UCSB)

- Developed an LSTM-based predictor for network throughput enabling closed-loop control under non-stationary dynamics.
- Designed a model predictive control (MPC) framework using receding horizon optimization for adaptive decision-making over bandwidth constraints.

In Motion — HAR using DeepConvLSTM

Nov 2019

Embedded Sensor Inference System for Temporal Activity Recognition (IIT Kharagpur)

- Designed a streaming inference pipeline for high-frequency IMU signals enabling real-time temporal classification.
- Implemented a DeepConvLSTM architecture combining convolutional feature extraction with temporal sequence modeling for activity recognition.

Battleship Solitaire Solver

Apr 2022 – Jun 2022

Solver-Aided Constraint Synthesis for Structured Search Problems (UCSB)

- Formulated Battleship Solitaire as a constraint-satisfaction problem in Rosette/Racket, representing grid cells and ship placements as symbolic variables subject to row/column tally, fleet-composition, adjacency, and revealed-cell constraints.
- Designed a solver-aided synthesis pipeline that parses puzzle instances, applies constraints as assertions, and uses Rosette’s angelic execution to synthesize concrete board configurations satisfying all constraints.

Industry Experience

Intuit Inc

Mountain View, CA · Mar 2024 – Jan 2026

Software Engineer 2

- Built **dual-stage agentic decision pipelines** using tool-schema routing for real-time intent classification and sequential state tracking.
- Developed a decoupled **generative UI engine** leveraging RAG and structured LLMs to optimize deterministic component layouts.

Yahoo Inc

Mountain View, CA · Apr 2023 – Nov 2023

Software Dev Engineer II (IC3)

- Developed **client-side rule engines** for asynchronous receipt payloads, executing deterministic filtering for **real-time decision support**.

Coupang Global LLC

Mountain View, CA · Jun 2022 – Sep 2022

Software Engineering Intern

- Engineered **context-aware heuristics** to prune multi-dimensional search spaces based on active user state vectors.
- Designed **Content-Based Filtering (CBF)** layers using **cosine similarity** to rank high-dimensional product features.

Samsung Research Institute

Noida, India · Sep 2020 – Sep 2021

Software Engineer (CL 2)

- Architected a **continuous on-device perception pipeline** ingesting multi-modal media streams for automated ML classification.
- Designed a **cross-profile data brokering engine** leveraging multi-user storage isolation to route assets from **User 1** to sandboxed **User 0**.

Teaching

University of California, Santa Barbara

Santa Barbara, CA

Graduate Teaching Assistant

Sep 2021 – Jun 2023

- Directed instructional labs, formulated algorithmic programming assignments, and monitored system architecture milestones for senior capstone teams.
- Covered advanced and core tracks: CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Technical Skills

Programming: Python, C++, Java, Kotlin, MATLAB

Robotics & Simulation: ROS/ROS2, Gazebo, RViz, Unity

Machine Learning & Vision: PyTorch, OpenCV

Methods: State Estimation (EKF), Model Predictive Control (MPC), Motion Planning, Sensor Fusion, Spatial Perception

Developer Tools: Git

Academic Service & Peer Review

Peer Reviewer, UIST 2024

ACM Symposium on User Interface Software and Technology

- Reviewed submissions on interactive systems, spatial tracking, and interface hardware.